

CITS5508 Practice Exam — Chapter 9

Unsupervised Learning Techniques

- This practice paper has **5 questions** for **60 marks**.
 - Allow approximately **two minutes per mark** (≈ 2 hours total).
 - Only UWA-approved calculators permitted.
 - Answers are in the separate file *Practice Exam - Chapter 9 (Answer Key).md*.
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Question 1. (12 marks)

- (a) Define **clustering**, and explain how the meaning of an instance's "label" differs between clustering and classification. (3 marks)
- (b) List **four** distinct applications of clustering. (4 marks)
- (c) Describe the **k-means algorithm** as a loop of two repeating steps. Why is it guaranteed to converge, and why might it converge to a **suboptimal** solution? (5 marks)
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Question 2. (12 marks)

- (a) Explain why **inertia** is **not** a good metric for choosing the number of clusters k . (3 marks)
- (b) Describe the **elbow method** and the **silhouette score** as two techniques for choosing k . (4 marks)
- (c) An instance has a mean intra-cluster distance $\mathbf{a} = 0.5$ and a mean nearest-cluster distance $\mathbf{b} = 2.0$. Compute its **silhouette coefficient**, and interpret what the value tells you. State the range of possible values. (3 marks)
- (d) State **two** situations in which k-means performs poorly. (2 marks)
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Question 3. (12 marks)

- (a) Describe how **DBSCAN** defines clusters. In your answer define **ϵ -neighbourhood**, **core instance**, and how anomalies are identified. (5 marks)
- (b) State **two** advantages of DBSCAN over k-means, and **one** limitation. (4 marks)
- (c) DBSCAN has no `predict()` method. Describe how you could classify a new instance into one of the clusters DBSCAN found. (3 marks)
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Question 4. (12 marks)

- (a) Describe what a **Gaussian Mixture Model (GMM)** assumes about how the data was generated. (3 marks)
- (b) GMMs are trained with the **Expectation-Maximization (EM)** algorithm. Describe the two steps, and explain how EM differs from k-means (hint: hard vs soft assignment, what extra properties of clusters EM learns). (5 marks)
- (c) Inertia and the silhouette score are not reliable for choosing the number of clusters in a GMM. Name **two** criteria you would use instead, and name a model variant that selects the number of clusters automatically. (4 marks)
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Question 5. (12 marks)

- (a)** Explain the difference between **anomaly detection** and **novelty detection**, focusing on what each assumes about the **training set**. *(4 marks)*
- (b)** Explain how a trained **GMM can be used for anomaly detection**, including how you would choose the density threshold and how it relates to the precision/recall trade-off. *(4 marks)*
- (c)** Name **two** dedicated algorithms (other than GMM) for anomaly/novelty detection, and in one sentence each describe how they identify anomalies. *(4 marks)*